

PhytoGraph — cycles 33-35

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PhytoGraph — cycles 33-35

Abstract

Cycles 33-35 closed the PhytoGraph free-tier recovery sequence at the master-ledger level. Cycle 33 produced the final six-track free-tier closure synthesis, a canonical status table, and a summary figure. The auditor validated that package and confirmed that no track met the predicate required for promotion into the master prediction_ledger.tsv or speculation_ledger.tsv.

Cycles 34-35 did not add substantive scientific work in the available record. The supplied audit report treats them as terminal no-op closure cycles: no worker build, no new artifact creation, no same-axis free-tier rerun, no promise-ledger append, and no master-ledger promotion occurred. The result is therefore a validated non-promotion state, not a new biological discovery.

The final track statuses are: Track 1 sidecar_readiness_uncontrolled, Track 2 H2_remains_not_supported_or_data_limited, Track 3 confound_limited, Track 4 still_data_limited, Track 5 H5_remains_source_biased, and Track 6 environment_limited_untested.

Introduction

PhytoGraph is a six-track plant-biology hypergraph campaign. Its original success criterion was not simply to build a dataset or website, but to produce falsifiable predictions and validate or falsify them cleanly. The master prediction ledger is therefore the controlling boundary: a track-local candidate, diagnostic, or prior does not become a campaign-level prediction unless it satisfies the recorded validation and promotion contract.

The cycles reported here follow earlier Barrier 4 and reopen work. By cycle 33, the campaign had already produced a large accepted-key substrate, track-local instruments, an Atlas surface, validation tables, ablations, and several closure reports. The remaining task was to reconcile the strongest free-tier outcome for each track without over-promoting branch-local hypotheses.

Approach

This report consolidates completed work rather than re-auditing it. The source inventory came from local workspace artifacts, the supplied audit report, and the supplied cycle-session identifiers. Native session-fetch tools were not available, so full researcher, worker, and auditor transcripts for the listed session IDs could not be fetched. That is recorded as a source gap.

The main sources were:

- `promise_ledger.jsonl`, especially cycle 33 worker event `3d046e5d-0f4d-4d37-9fd9-2e499032455f` and auditor event `9910b77f-f139-43fa-b0e1-b3ffa297c050`.
- `reports/reopen/final_free_tier_closure_synthesis.md`.
- `data/reopen/final_free_tier_track_status.tsv`.
- `reports/reopen/figures/final_free_tier_track_status.png`.
- `Root final_report.md` and `audit_report.md`.
- The supplied audit report for cycles 33-35.
- `prediction_ledger.tsv`, `speculation_ledger.tsv`, and `MANIFEST.md`.

Findings

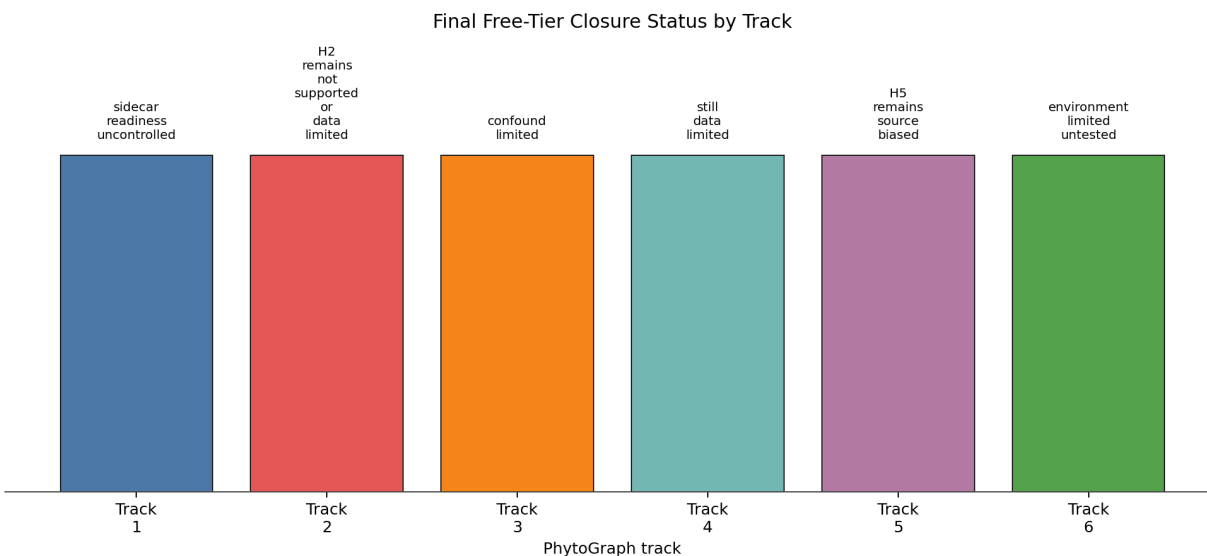
Cycle 33: Final Free-Tier Closure Synthesis

Cycle 33 produced the final free-tier closure synthesis across all six tracks. The worker event states that the synthesis was produced without new evidence search, schema change, branch-science change, or master-ledger promotion. The auditor then validated the package after inspecting the status TSV, synthesis report, generator/test, root and reopen report propagation, summary figure, worker ledger event, prediction/speculation ledgers, and focused tests.

The canonical cycle 33 output was `data/reopen/final_free_tier_track_status.tsv`, with one row per track:

Track	Final status	Key counts	Claim boundary
Track 1 Reticulation Atlas	<code>sidecar_readiness_uncontrolled</code>	22 GBIF event taxa; 11 source groups; 2 WFO-projected taxa; 0/17 matched-control event recovery	No established reticulation hotspot, hybridization, or polyploid recovery claim
Track 2 Ghost Hyperedges	<code>H2_remains_not_supported_or_data_limited</code>	8 canonical held-outs; 31 local candidates; 0/8 held-outs pass validation contract	No new anachronism, ghost-partner, or ecological-interaction claim
Track 3 Convergence Pressure	<code>confound_limited</code>	3,069 accepted-key trait carrier rows; 15 canonical traits; 0 controlled-ready traits	No convergence or adaptive-origin claim

Track	Final status	Key counts	Claim boundary
Track 4 Domestication Hypergraph	still_data_limited	3,358 post-filter occurrence records; 0 numeric BIOCLIM vectors; 0 validation-allowed comparator rows	No crop-substitution recommendation or climate-adaptation claim
Track 5 Chemodiversity Predictor	H5_remains_source_ biased	Non-Duke temporal evidence insufficient; no validation-ready structured family/class stratum	No new phytochemical, bioactivity, or screening-priority claim
Track 6 Foundation Model Probe	environment_limited_ untested	0 runnable runtime-weight pairings; 0 executed responses; 0 scored responses	No model error-rate, leaderboard, toxicity-look-alike, or vendor-comparison claim



Final free-tier closure status for all six PhytoGraph tracks, distinguishing sidecar-readiness, not-supported, confound-limited, data-limited, source-biased, and environment-li

Figure 1: Final free-tier closure status for all six PhytoGraph tracks. The figure distinguishes sidecar-readiness, not-supported, confound-limited, data-limited, source-biased, and environment-limited closure states.

The synthesis makes the master-ledger decision explicit: `prediction_ledger.tsv` and `speculation_ledger.tsv` remain header-only because every track retains at least one failed validation predicate, unresolved control, source-bias collapse, or execution blocker.

Cycle 34: No New Substantive Artifact Found

No local cycle 34 branch report, worker artifact, validation table, figure, test, or ledger append was found during the workspace inventory. The supplied audit report also records no new worker build, no artifact creation, and no master-ledger promotion.

The correct reportable result for cycle 34 is continuity of the cycle 33 terminal closure state. No scientific claim changed.

Cycle 35: Terminal No-Op Closure Confirmed

Cycle 35 is represented in the supplied audit report as a terminal-state assessment rather than a build. The audit decision is `VALIDATED`, with zero critical, moderate, or minor findings.

The supplied audit validated these invariants:

- Final free-tier closure synthesis remains auditor-validated and closed.
- Barrier 1, Barrier 2, Barrier 3, Barrier 4, M5.1, reopen branches, closure addenda, and final synthesis remain auditor-validated per the supplied ledger/context.
- `prediction_ledger.tsv` remains header-only with 1 line.
- `speculation_ledger.tsv` remains header-only with 1 line.
- `promise_ledger.jsonl` remains at 226 events.
- No worker build, artifact creation, same-axis free-tier rerun, or master-ledger promotion occurred.

The audit guidance states that the campaign should be treated as terminally closed at the master-ledger level. Reopening requires new qualifying evidence that satisfies a recorded track predicate, such as controlled WFO-projectable reticulation evidence, Track 2 held-out multi-source support, controlled-ready convergence traits, observed climate vectors with comparator rows, non-Duke dated chemistry strata, or audited local/open model responses.

Discussion

Cycles 33-35 did not create a new discovery. They converted the prior branch-level closure work into a final, auditor-validated non-promotion state.

That distinction matters. PhytoGraph built substantial infrastructure: a typed hypergraph substrate, track-local instruments, Atlas integration, validation and ablation packages, and formal diagnostics. But the campaign's original success criterion was stricter than infrastructure completion. A track had to produce a falsifiable prediction and clear its validation protocol. By the end of these cycles, no track had cleared that standard at the master-ledger level.

The final synthesis also prevents repeated null work. Each track now has a specific reopen predicate. Future work should vary the missing evidence axis rather than rerunning the same free-tier branch inputs.

Open Questions

The remaining questions are framed as reopen predicates rather than open-ended research invitations:

- Can Track 1 map GBIF accepted-key sidecar evidence into the frozen WFO accepted-key namespace, or admit an audited sidecar namespace with source-density controls?
- Can Track 2 obtain accepted-key modern-failure evidence and multi-source/source-class support for held-out anachronism cases?

- Can Track 3 obtain broader trait coverage and controls that separate convergence from family size, sampling density, projection loss, and source dominance?
- Can Track 4 generate audited crop/CWR BIOCLIM summaries plus disjoint candidate-level expert comparator rows?
- Can Track 5 obtain enough dated, non-Duke taxon-compound rows to estimate chemistry signatures without source collapse?
- Can Track 6 run approved local/open model weights and produce audited deterministic response rows?

References

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- [31] Daniel Falster, Rachael Gallagher, Elizabeth Wenk, Herve Sauquet, et al., “AusTraits: a curated plant trait database for the Australian flora,” Zenodo, version 6.0.0, 2024. <https://doi.org/10.5281/zenodo.11188867>.
- [38] James A. Duke, “Dr. Duke’s Phytochemical and Ethnobotanical Databases,” Ag Data Commons, 2023. https://agdatacommons.nal.usda.gov/articles/dataset/Dr_Duke_s_Phytochemical_and_Ethnobotanical_Databases/24660351.

Appendix: Implementation Details

Source Inventory

Source	Date / cycle	Contents	Timeline role
Cycle 33 worker session 46bd1b76-8178-4787-8974-9b8e897d4251	Cycle 33	Listed in supplied cycle_sessions; full transcript unavailable locally	Worker for final closure synthesis
Cycle 33 auditor session 43409405-ba54-4fa4-a1ec-9746e6eac845	Cycle 33	Listed in supplied cycle_sessions; full transcript unavailable locally	Auditor for final closure synthesis
promise_ledger.jsonl event 3d046e5d-0f4d-4d37-9fd9-2e499032455f	2026-05-18 / cycle 33	Worker registration for final free-tier closure synthesis	Created final synthesis package
promise_ledger.jsonl event 9910b77f-f139-43fa-b0e1-b3ffa297c050	2026-05-18 / cycle 33	Auditor validation for final free-tier closure synthesis	Superseded worker event and validated non-promotion

Source	Date / cycle	Contents	Timeline role
reports/reopen/final_free_tier_closure_synthesis.md	Cycle 33	Final closure narrative and six-track status table	Main report source
data/reopen/final_free_tier_track_status.tsv	Cycle 33	Machine-readable final status table	Canonical status source
reports/reopen/figures/final_free_tier_track_status.png	Cycle 33	Summary figure	Embedded evidence figure
Supplied audit report	Cycles 33-35	Terminal no-op validation and invariant list	Source for cycles 34-35 no-op closure
Cycle 34 sessions 51a701a4-99e8-4984-a11b-cef2a0e16fdc, 7da378c7-f829-4bd9-b437-c76eec96780e, b5dc95c0-6240-4ace-811c-bc1451c0a3d7	Cycle 34	Listed in supplied cycle_sessions; full transcripts unavailable locally	No local substantive artifact found
Cycle 35 sessions 5dea5728-a372-4971-abc6-64f9968ea803, 3079f49d-e3c1-4cb7-adb1-4ce74672b696, fd1060f2-bc7a-4ca6-b094-da24e9475e45	Cycle 35	Listed in supplied cycle_sessions; full transcripts unavailable locally	Supplied audit records terminal no-op validation

Code Organization

File	Lines	Purpose
reports/reopen/scripts/build_final_free_tier_closure_synthesis.py	197	Builds the final six-track free-tier status table, closure report, and summary figure.
tests/test_final_free_tier_closure_synthesis.py	109	Verifies final status table, root/reopen propagation, figure presence, and header-only master ledgers.
tests/test_reopen_closure_addendum.py	70	Verifies reopen closure status coverage, future-data predicates, forbidden-claim boundaries, and master-ledger non-promotion.

Data And Report Files

File	Rows / lines	Role
reports/reopen/final_free_tier_closure_synthesis.md	45 lines	Final six-track closure synthesis.
data/reopen/final_free_tier_track_status.tsv	6 data rows	Canonical final status table.
final_report.md	128 lines	Root final report with final free-tier closure position.
audit_report.md	108 lines	Root audit report with terminal closure judgment.
research_contribution_ledger.md	58 lines	Contribution ledger preserving non-promotion classification.
falsification_and_ablation_report.md	139 lines	Falsification and ablation closure report.
artifact_index.md	160 lines	Artifact map for final/reopen packages.
prediction_ledger.tsv	1 line	Header-only master prediction ledger.
speculation_ledger.tsv	1 line	Header-only master speculation ledger.
promise_ledger.jsonl	226 events	Append-only campaign ledger.

Test Results

The reporter did not rerun tests. The supplied audit report records:

- `python3 -m long_exposure.tools.promise_check /home/jeremyad55/workspace:` exit 0, 226 events, inherited warnings only.
- `python3 -m long_exposure.tools.org_check /home/jeremyad55/workspace:` exit 0, inherited root-layout warnings only.

The cycle 33 auditor event also records `pytest: 42_passed`, figure check `nonblank_1980x990`, and no schema change, branch-science change, external fetch, or master promotion.

Cross-Reference Map

Origin	Consumer	Flow
Cycles 30-32 free-tier branch packages	Cycle 33 final synthesis	Earlier Track 1, Track 2, and Track 3 closure outcomes were consolidated with Track 4, Track 5, and Track 6 closure states.
data/reopen/final_free_tier_track_status.tsv	reports/reopen/final_free_tier_closure_synthesis.md	Supplies the canonical status, blocker, future-data requirement, and claim boundary for each track.

Origin	Consumer	Flow
Cycle 33 worker event	Cycle 33 auditor event	Auditor validated and superseded the worker registration.
Supplied cycles 33-35 audit report	This periodic report	Establishes cycles 34-35 as no-op terminal closure.
prediction_ledger.tsv and speculation_ledger.tsv	All cycle 33-35 conclusions	Header-only state is the controlling non-promotion decision.
MANIFEST.md	Future researcher/worker turns	Updated as a concise snapshot for cycles 33-35 while preserving the existing ## Key Files section verbatim.

Record Gaps

Full local transcripts for the supplied session IDs were not available because session search/fetch tools were not available in this environment. The report therefore relies on local artifacts, ledger entries, and the supplied audit report.

No local cycle 34 or cycle 35 substantive artifact was found. This matches the supplied audit's no-op terminal assessment and is reported as absence of new work, not as a scientific result.

Coherence Review

One coherence pass was completed. The report defines the master-ledger promotion boundary before using it, distinguishes cycle 33 artifact creation from cycle 34-35 no-op closure, and marks the unavailable session transcripts as a record gap rather than inferring missing content.